Vocabulary position and trademark lifecycles:
An event-dated corpus and a lead/lag text measure for the commercial economy

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July 2026 working paper

Abstract

Text-based novelty measures built on patents observe invention by the small minority of firms that patent. Trademark filings time-stamp commercialization by everyone else, in plain commercial language.

All 13.99 million USPTO trademark case files are re-parsed into event-dated prosecution histories, and each filing scored on one idea: how surprising its language is. The question is put to two readers with disjoint knowledge — one who has seen only what the industry filed before that filing, one who has seen only what it filed after. *Atypicality* is how surprising the wording is to them on average. *Lead*, written ΔKL, is the gap between them: positive when the wording surprised the first reader and not the second, which is what it looks like to have come first. Such filings are *leading*; those whose language the industry was already leaving behind are *lagging*.

Among marks the office granted, it is *timing* rather than unusualness that predicts what happens next. Leading marks — those whose language their industry subsequently moved toward — fail their first use-proof gate — the declaration, due between the fifth and sixth year, that the mark is still in commercial use — 1.4 percentage points more often in the top within-class-cohort quintile than in the bottom, about 53.3% against 51.9% on a 52.3% base failure rate, on 3.10 million registrations with class×cohort fixed effects, owner-clustered errors, and drafting and filer controls ($t = 8.3$). The penalty is indistinguishable from zero in the earliest cohorts, present in every cohort from 2008 to 2018, and reproduces out of sample in 2019; almost all of it sits in the top quintile rather than accumulating across the range. Leading debut owners also become SEC-reporting companies less often, -0.047pp per standard deviation ($t = -7.8$); about half of those firms are exchange-listed and the rest report for other reasons, so this is a coarser marker than an IPO. Atypicality — how unusual the wording is, without regard to direction — does none of this work: its association with listing is indistinguishable from zero once the owner-to-registrant linkage spans the whole corpus, and it is sensitive to how that linkage is drawn in a way the signed axis is not. The measure exists for the millions of firms that never patent and is essentially uncorrelated with patenting within firm.

Two cautions bound how far this travels. Read without conditioning on grant, the same data support weaker and sometimes opposite conclusions, because registration is itself selected on language. And scored on individual terms rather than topic distributions, leading filings appear to belong to better firms — an association that dissolves or reverses under topic scoring, because term scoring picks up lexical specificity rather than position. Text measures of this kind can manufacture firm-performance correlations out of drafting style. Code, panels, and the corpus build are public at `https://github.com/ericsilver/tm-vocabulary`.

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1 Introduction

Text-based measurement has reshaped how economists quantify invention. Kelly et al. (2021) score every US patent since 1840 by its textual distance from prior and similarity to subsequent patents; Kalyani (2024) measures patent creativity as the share of technical terminology not seen before and ties it to firm productivity. Both programs inherit the patent record's coverage: technical invention by the small minority of firms that patent. Most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer the complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year.

The trademark–innovation literature is three decades old and has moved well past counting. Flikkema et al. (2025) survey its first ten years of validation work and its open questions; Willeke et al. (2026) inventory the data sources and the preprocessing burden that has kept them underused. Micro-level links to firm dynamics are established (Dinlersoz et al., 2018, 2023), and the goods/services description is already read as text rather than treated as a label: Semadeni (2006) scores consulting service-mark descriptions into a competitive positioning space, Semadeni & Anderson (2010) tracks whether one firm's described offering is imitated by rivals, and Flikkema et al. (2019) codes descriptions against the innovation each mark accompanies. Prosecution records have been exploited too: Fink et al. (2024) use contested filing sequences to study trademark races, and their outcome — continued use five years after registration — is the margin this paper's central result also runs on. What Flikkema et al. (2025) put as the field's next step is the move from counts and innovation intensity to the *type*, *quality*, and *trend* content of what marks cover: the trademark analogue of what Kelly et al. (2021) and Kalyani (2024) did for patent text. That analogue does not yet exist at corpus scale, and the two strands above — description text and prosecution events — have run on separate data.

This paper builds both on one corpus, so that what a filing says and what subsequently happened to it are observed on the same record. The first contribution is that corpus. The USPTO distributes its trademark backfile as bulk XML — 83 archives covering 1884 to 2025 — from which the published research files retain a filing's current status but not the dated sequence of events that produced it. Re-parsing the backfile yields 242 million prosecution events — every step in the examination of an application — across 13.99 million case files, each event carrying a code and a date: every office action, response, maintenance filing, and cancellation. Three further passes make those events usable. The event codes are resolved into a 760-code dictionary. A header pass extracts counsel of record, the statutory filing basis, and the postregistration declarations. A third pass recovers owner domicile, which the class-level files omit. Owner names are then normalized to a clustering key and resolved to two external universes: SEC registrants, through the agency's own company-name and ticker files, and Regulation D issuers, through the Form D quarterly data sets — the latter filtered to operating companies, with ambiguous names dropped rather than guessed. That link is what lets a mark's language be followed to financing and to public listing. Where the patent record dates invention once (at grant), the trademark prosecution record dates a commercial claim's entire life, including the use-proof gates at which marks attached to products that never materialized are culled. The second contribution is a filing-level text measure defined on that corpus. The third is a measurement hazard, documented at corpus scale, that applies to text-based novelty measures generally.

The procedure measures one thing: how surprising a filing's language is. Group every trademark filing by its industry class — one of the 45 international goods and services categories — and place it by its filing date, then put the filing to two readers who have seen disjoint halves of that industry's stream: one who knows only what the class filed before it, one who knows only what it filed after. Each reader's "how surprising" is a Kullback–Leibler divergence, the standard measure of how unexpected one distribution looks from another, between the filing's mix of topics — its weights over word-clusters fitted to the corpus — and the industry's. It is large when the filing puts weight where its industry's filings put little. *Atypicality* is the average of their two answers. *Lead*, written ΔKL, is the gap between them.

What the gap captures is worth stating plainly. Language that surprised the first reader and not the second was surprising because it came first: the industry moved toward it. Language that read as ordinary to the first and unusual to the second was behind its industry rather than ahead of it, and was left behind as the field moved on. Filings of the first kind are *leading*, of the second *lagging*. So atypicality asks whether a filing said something surprising, and lead asks whether the surprise was merely a head start. Both are computable for every filing in the corpus, including the millions from firms that never patent.

Trademark law then supplies an unusually direct test, because a registered mark must be re-proved in use five to six years on or be cancelled — a dated, adversarial checkpoint on whether the product the mark named ever reached the market.

The construct adapts the prospective/retrospective Kullback–Leibler framework that Murdock et al. (2017) developed on Darwin's reading notebooks and Barron et al. (2018) extended to French Revolution parliamentary debates. The nearest measurement precedent on this corpus is von Graevenitz et al. (2022), who flag goods/services tokens that are both new and fast-spreading and use their geography to show that distance still retards diffusion; their unit is the token and their flag is binary, while the score here is continuous and attaches to the filing, so every application carries a position that can be crossed with the mark's own prosecution outcomes. Section 3 sets out what transfers from that literature and what does not — chiefly that there is no single authoring agent here, and that legal drafting rewards caution about convention, so what looks like surprise may be style, length, or counsel effects. That is why the design leans on fixed effects, drafting covariates, and owner clustering rather than raw gradients.

The central results are stated on marks the office actually granted, and that conditioning is not incidental. Registration is itself selected on language, so a correlation computed over all filings mixes what the examination process does to a description with what the market does to the product. Appendix C reports the unconditional version and shows where it diverges.

Among granted marks, the signed axis — not the unsigned one — predicts what happens next. Debut filings whose language their industry subsequently adopted belong disproportionately to owners who never appear in SEC financial-reporting records. Conditional on being surprising, having been early is what separates the failures. Leading marks fail the use-proof gates more often, and at equal atypicality their owners list less often, not more. Having been early, in other words, is the dimension along which surprise stops paying. The risk side survives a design built to kill it — fixed effects, drafting and filer controls, owner clustering, and estimation inside the counsel-represented US-domiciled stratum.

Computation details, scoring robustness, and the measurement-hazard forensics are in the appendices.

2 Registration measures follow-through; the use-proof gate measures survival

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a statement of use is filed, the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a use-proof gate is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific gate by its date rather than inferred from a status snapshot.

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; gate outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Sources: the TRTYRAP re-parse and the scored filing panels described in Section 3.

Stage	n	Share
Debut filings scored, 1995–2018 (owners' first filings)	3,589,068	
reached registration	2,066,600	57.6%
of failures: never filed statement of use		42.6%
of failures: stopped responding in prosecution		45.3%
of failures: removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the first use-proof gate (event-dated)		52.3%
Passed first gate, cohorts 2002–2013, terminal status known	1,129,724	
failed at the year-ten renewal		39.1%

Three windows bound the samples, and each is set by an institutional fact rather than by preference. Scoring runs 1995–2019, because a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020; the analysis cut of 1995 is about two years more conservative than the measured burn-in requires (Appendix A). The gate cohorts end in 2018 because the cancellation record runs to April 2026 and the failure window closes at registration age 8.5 years, so 2018 is the last cohort whose window has essentially elapsed — the top of that range is fractionally censored, and a separate 2019 replication is reported below. They begin in 2002 for a weaker reason: it is a conventional cut that predates any of the reported results, and the 2002–2007 third is in any case the one third in which the penalty is smallest. Nothing in the paper turns on the lower bound.

The funnel (Table 1) fixes the paper's reading of each step. Registration is a launch indicator rather than an examiner verdict: among applications that fail to register, roughly nine in ten die by the applicant's own inaction — never filing the statement of use that registration requires, or going silent during prosecution — and only 4.3% are removed by opposition, appeal, or other adversarial proceedings. Completing registration therefore mostly measures whether the filer followed through to actual commercial use. The Section 8 gate is where the market's verdict reaches the record five years later: half of registered marks do not survive it, and failing it means the product or service the mark named is no longer (or never was) in commerce.

All lifecycle results below condition on this funnel once. The registration step is analyzed as its own outcome; the gate results condition on registration, so they compare marks that launched; and the unconditional composite — which mechanically stamps the registration step's shape onto any downstream outcome — is confined to Appendix C. Outcomes are read two ways throughout: from the USPTO status-code dictionary,[1] ground-truthed against marks with publicly known fates, and, for the primary results, from the event-dated prosecution history, which attributes each cancellation to a specific gate and observes counsel and basis directly.

3 The construct: how surprising a description is, and when

3.1 Two questions asked of every filing

The two readers of the introduction can now be made precise. Take a filing's goods/services description and group every filing by its Nice class — one of the 45 international goods and services categories. Each reader has seen only half of that class's stream, split at the filing's own date:

1. A reader who has seen only what the industry filed *before* it. How surprising is the wording to them?
2. A reader who has seen only what the industry filed *after* it. How surprising is the wording to them?

Neither reader knows the other's half; the two references are disjoint. Each window is anchored on the filing's own date — the 1,826 days before it and the 1,826 days after it — so no two filings made on different days are scored against the same reference, and filings made on the same day are excluded from each other's windows. Within a window every day counts the same: the reference is a flat pooled aggregate, not a decayed one, so a term the class adopted five years ago and one it adopted last year are equally familiar to the first reader. The measure therefore asks whether the wording is unusual against a period, not whether it is ahead of a trend within that period; it has no momentum channel, and Section 6 returns to what that forecloses.

Anchoring on the filing date rather than on the calendar year is not a detail. The alternative — one reference per class-year — scores a filing made in January and one made in December of the same year against the same object, and lets neither see any of the language filed during its own year. That imprints a spurious gradient: under annual buckets mean lead rises almost monotonically with filing month, purely because a December filing sits eleven months further from its past reference and eleven months nearer its future one. Appendix A.5 reports what the coarser version cost, which was roughly a third of the measured gate penalty and the whole of an apparent curvature at registration.

The two answers are usually close, and the gap between them is what carries the information. On the software class they correlate at 0.979, and the difference between them has about a sixth of the spread of either level. Wording that surprised the first reader and not the second was surprising because it came first: the industry moved toward it. Wording that read as ordinary to the first and unusual to the second was behind its industry rather than ahead of it, and was left behind as the field moved on. Filings of the first kind are *leading*; of the second, *lagging*. That the signal is a small residual of two large and highly co-moving quantities is a real fragility rather than an aside, and Appendix A quantifies it.

Both quantities are computable for every filing in the corpus, including the millions from firms that never patent, and neither requires knowing anything about the firm.

[1] 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

One property of the underlying text shapes how the resulting numbers should be read. Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language — from the USPTO's own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors' successful filings. Identical wording across unrelated firms is therefore common rather than anomalous, though under per-filing reference windows an exact tie in the resulting scores is rare (Section 4.2). Two consequences follow. Within a class, the measure is picking up departure from a shared drafting convention, which is what makes it meaningful. Across classes, it is not: a class where the Manual supplies dense standard language will show a compressed distribution of atypicality, and one where firms describe novel services in prose will show a wide one, for reasons that have nothing to do with how innovative either industry is. Every estimate in this paper is therefore taken within class and year, and the magnitudes should be compared across classes only as signs and orderings, never as levels.

3.2 Antecedents, and what does not carry over

The apparatus is that of Murdock et al. (2017), developed on Darwin's reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score each document against the distribution of documents before it and again against the distribution after it, and call the two quantities *novelty* and *transience* and their signed difference *resonance*.

One methodological choice is inherited rather than invented. Both source papers score documents on their distribution over latent topics rather than on raw terms, and this paper reports topic scoring throughout, with term scoring retained as a cross-check.[2]

The two representations measure different things, and neither dominates. Under term scoring, atypicality correlates with log filing length at -0.729 in class 009 and -0.688 in class 035; under topic scoring, at -0.251 and -0.125. Descriptions run to a few dozen words, and on documents that short a term-level divergence rises as the text gets shorter, so "unusual" and "brief" are close to the same measurement. Correspondingly, the spread of the signed difference across levels of the unsigned average — written L and A once they are defined below — varies roughly two- to threefold under term scoring against 1.3- to 1.6-fold under topic scoring (Appendix A). In the other direction, Appendix E takes three filings whose novelty is known independently — a combinatorial one, a lexical one, and one riding a wave — and finds that term scoring resolves all three while topic scoring resolves none at any resolution tested, because a topic basis fitted on the corpus has no coordinate for language the corpus has not yet seen. Every worked example in this paper is accordingly term-scored, and every estimate is topic-scored. A reader who wants the headline results under term scoring will find them in Appendix A.

The arithmetic transfers exactly. The interpretation does not, in two ways that shape everything downstream.

First, there is no single authoring agent. In a reading notebook the author of the document is the person whose novelty is being measured, and the reference stream is that author's own intellectual environment; a high value therefore says something about a mind. Here each filing is scored against the stream of *other* firms' filings, and the author is typically counsel drafting for scope of protection. The same number can only say that a description sits far from what its industry was writing.

Second, *resonance* names a mechanism this design cannot observe, for reasons Section 6 sets

[2] I am grateful to Simon DeDeo for pointing out that the topic operationalization, rather than term-level scoring, is the one his co-authors used, and for suggesting it be applied here. The consequences were larger than a robustness check: Appendix B reports a set of firm-level correlations that survive term scoring and dissolve or reverse under distributions.

out. The vocabulary used below is therefore positional rather than causal: a leading filing was ahead of its industry's language, which implies the existence of a trajectory without asserting that the filing caused it.

3.3 Definitions

Let P_i be the topic distribution of filing i, made on date d_i, and let Q_{past} and Q_{future} aggregate same-class filings over the W days strictly before and strictly after that date, $[d_i - W, d_i)$ and $(d_i, d_i + W]$. The two windows are disjoint, neither contains the filing itself, and filings sharing its date fall in neither. $W = 1{,}826$ days (five years) throughout unless stated; the window is a parameter of the measure rather than a property of it, and Appendix A reports what changes when the two windows are set to different lengths. Write

$$K^- \;=\; \mathrm{KL}(P_t \,\|\, Q_{\text{past}})\,, \qquad K^+ \;=\; \mathrm{KL}(P_t \,\|\, Q_{\text{future}})\,.$$

Kullback–Leibler divergence is large when the filing puts weight where its industry's filings put little, so K^- is surprise measured against the past and K^+ surprise measured against the future. The source literature and the appendices below call K^- *prospective* and K^+ *retrospective* KL; those names describe the reader's vantage rather than the direction of the window, so prospective surprise is surprise measured against what came before. The two axes used throughout are

$$\underbrace{A \;=\; \tfrac{1}{2}\left(K^- + K^+\right)}_{\text{atypicality}}, \qquad \underbrace{L \;=\; K^- - K^+}_{\text{lead}}.$$

A is how unusual the wording is overall, in both temporal directions at once. L is signed: positive for a leading filing, negative for a lagging one. The symbol $\Delta\mathrm{KL}$ is used interchangeably with L where it aids comparison with the source literature.

3.4 Why these two, and not the raw pair

A and L are a rotation of (K^-, K^+) through 45 degrees. Plotted with K^- on one axis and K^+ on the other, A is position *along* the diagonal and L is signed distance *from* it: magnitude and direction of the same displacement. Three things follow.

The rotation is the natural coordinate system because the raw pair is almost collinear. K^- and K^+ correlate at 0.979 on the complete software-class corpus, so entering both is close to entering one variable twice; the informative content is the long axis and the small residual, which is exactly what A and L isolate.

The rotation also very nearly decorrelates the two regressors, which matters because the paper enters both and interprets them separately. Against L, the correlation of A is $+0.038$; of K^- alone, $+0.140$; of $|L|$, $+0.250$ ($n = 1{,}107{,}807$). Using either level alone, or the unsigned magnitude, as the "how unusual" axis leaves the two coefficients trading against each other. Nothing empirical rides on the choice — given the 0.979 correlation, A and K^- are nearly the same variable — but the interpretation is cleaner and the coefficients are closer to independent. Table 2 sets the five quantities against their names in the source literature.

3.5 A note on the words

Atypicality is used in a purely textual sense and is not trademark law's distinctiveness doctrine, which governs registrability along the generic-to-fanciful spectrum; the two attach to different objects, one to the mark and one to the description of the goods it covers. Section 6 separates the

Table 2: The quantities, their sources, and the terms used here. *Lagging* and *leading* describe the two signs of L; *atypicality* is unsigned and says nothing about direction.

Quantity	Definition	Murdock et al.	Term used here
Surprise against the past	K^-	novelty	past-facing surprise
Surprise against the future	K^+	transience	future-facing surprise
Their average	$A = \frac{1}{2}(K^- + K^+)$	—	**atypicality**
Their signed difference	$L = K^- - K^+$	resonance	**lead** (leading / lagging)
Its magnitude	$\lvert L \rvert$	—	lead magnitude

construct from its other near neighbours and states in each case what adopting that term would claim.

Leading and *lagging* were chosen over the alternatives for the same reason the causal reading was declined above. To say a filing led its industry's vocabulary is to place it relative to a trajectory without saying it produced one, in the way that being ahead of a curve implies the curve without implying you drew it.

4 Among granted marks, being early costs and being unusual does not pay

4.1 Why these results condition on grant

Registration is not a neutral filter on language. Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: at twenty bins it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 2, panel C). The same shape appears on the past-facing level alone, more mildly — 54.0% at the bottom quintile, 58.4–59.1% in the middle, 57.7% at the top (Figure 1A). Applicants whose language sits far from their category in either direction follow through to commercial use least often, and the penalty is larger for the conventional tail than the unusual one.

The signed axis behaves differently, and a five-bin summary of it misleads. At twenty bins, completion on ΔKL falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end (Figure 2, panel C). That trough is composition rather than an effect of timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. Reference windows built by calendar year rather than by filing date put an inverse-U on this axis too, but that one is an artifact of the annual bucketing documented in Appendix A.5 and does not survive at daily resolution. The signed axis still enters a linear specification strongly (Table 6), but it does so monotonically rather than through an interior peak: what curvature there is at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

That is a result about the examination funnel, not about the market. Pooling granted and abandoned filings mixes the two selections, and the resulting gradients are attenuated and in places reversed relative to the conditional ones; Appendix C reports that comparison in full.

It would be reasonable to expect unfamiliar language to be refused more often, and to suspect that the measure is therefore recovering nothing beyond familiarity. Two features of the shape argue otherwise. If unfamiliarity alone drove the outcome, completion would fall monotonically in the level of surprise; instead it falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain. And the curvature is specific to the unsigned axis: the signed axis, which is by construction almost uncorrelated with it, shows no interior peak at all (Figure 2, panel C), so whatever produces the shape is not simply distance from the category in the direction of travel. What the examination funnel appears to reward is neither novelty nor conventionality but a middling specificity — descriptions concrete enough to survive a definiteness objection and conventional enough to avoid a likelihood-of-confusion refusal. That is a claim about examination, and the paper does not build on it.

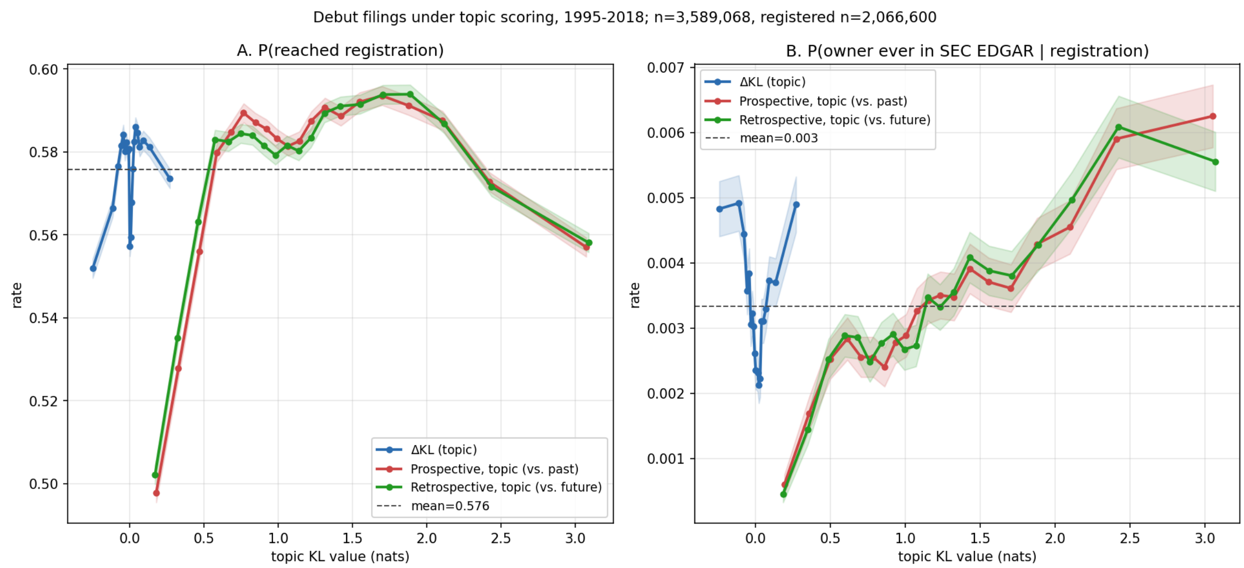

Figure 1: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under topic scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}068$; registered subset $n = 2{,}066{,}600$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR | registration); the structure in panel B is decomposed in Section 4.3.

4.2 Leading marks fail the use-proof gate

Marks written in language their industry subsequently moved toward fail their first use-proof gate more often than marks written in language it was moving away from. Within class×cohort cells, with owner-clustered errors and drafting and filer controls, marks in the top ΔKL quintile fail +1.4pp more often than those in the bottom ($t = 8.3$): percentage points on a 52.3% base rate, not percent of it, so roughly 51.9% against 53.3%, or about 2.7% more failure in relative terms. This is the paper's central result.

Almost all of it sits in the top quintile. Against the bottom quintile, the second through fifth run +0.15, +0.26, +0.42 and +1.39pp: the step from the fourth quintile to the fifth is seven times the average of the three below it (Figure 2, panel A). This is not a gradient in which each step of vocabulary position costs a little more, but a threshold in which the most strongly leading fifth of filings carries essentially the whole penalty. A coefficient reported per quintile would spread that concentration evenly across the range and misdescribe it; the estimates below are therefore top-versus-bottom contrasts throughout, and the profile is shown rather than summarized.

Two conventions govern this subsection. Estimates are on gate *failure* unless the text says survival; the window decomposition below and the appendix report survival, where the same penalty carries a negative sign. And the gate is the first maintenance filing, which is Section 8 for domestic

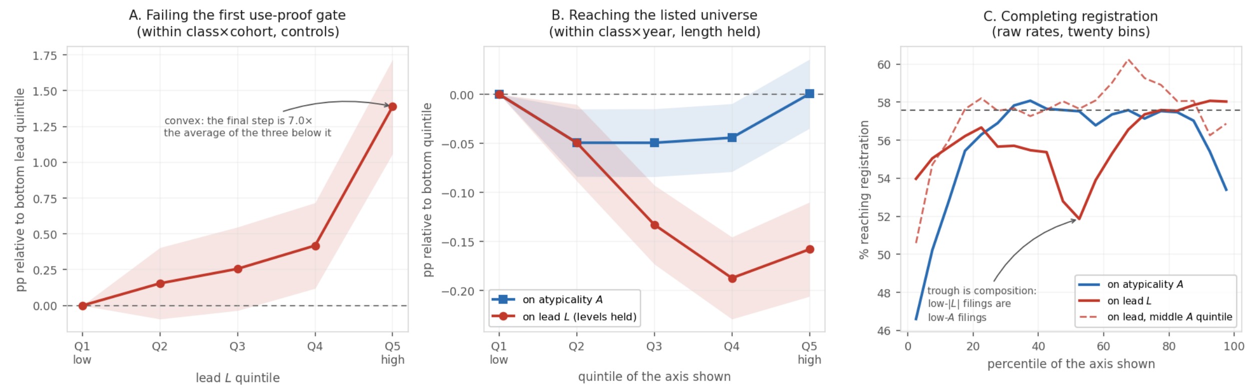

Figure 2: Quintile profiles for the three outcomes, on a common axis. Shaded bands are 95% intervals. *A:* first-gate failure against lead, within class×cohort with controls, relative to the omitted bottom quintile — convex, with the penalty concentrated in the top quintile. *B:* becoming SEC-reporting, within class×year with length held — flat in atypicality once length is absorbed, and falling in lead through the fourth quintile once the levels are controlled. *C:* raw registration rates for debut filings in twenty bins — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate. No parametric curve is fitted through five points: the claim is that these profiles are not lines, not that they are any particular other function.

registrations and Section 71 for Madrid Protocol Section 66(a) extensions of protection — the same gate under two statutes — §66(a) registrations are US extensions of an international registration filed through the Madrid Protocol — both due between the fifth and sixth anniversary. Both cancellation codes are read. Scoring only the Section 8 code, as earlier versions of this analysis did, records every Section 66(a) registration as a survivor whatever became of it, which is not a small distortion: those registrations then carry a spurious −46pp coefficient, and correcting it moves the same coefficient to +10pp, since Madrid registrations in fact fail this gate more often than domestic ones.

Raw pooled quintiles convey the direction: among 4.12 million registration class-records 2002–2018, dated first-gate cancellation rises from 50.0% at the bottom topic-ΔKL quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and cohort composition, treats multi-class registrations as independent observations, and ignores that gate outcomes correlate within owner. In other words, part of that raw gap is simply that industries and years differ in how often they file unusual language and in how often their marks fail, which is not the same thing as what unusual language does to a mark inside its own industry. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with ΔKL quintiles cut *within* class×cohort cells, a cohort here being a year of registration. The specification adds class×cohort fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;[3] and standard errors clustered on 1.27 million

[3] A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

normalized owners (Table 3).[4]

Table 3: First-gate failure and ΔKL: design-based estimates. LPM of event-dated first-gate failure on within-class×cohort ΔKL quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×cohort fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5−Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per σ	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per σ	+0.63	0.06	1,888,352

Standard errors are reported in place of significance stars. With n in the millions every coefficient in this table has $p < 0.01$, so conventional thresholds would mark each row identically and convey nothing; the standard error lets the reader form an interval and weigh the magnitude against the 52.3% base rate. Errors are clustered on *normalized owner* — owner names uppercased, stripped of punctuation and of twenty common legal suffixes, so that ACME CORP. and ACME CORPORATION are one cluster. Clustering there rather than on the filing is what the data structure requires: a single owner files many marks, drafts them in a house style, and abandons them together when the business fails, so filings within an owner are far from independent draws. The within-owner correlation of gate survival is 0.42. Treating 3.10 million filings as 3.10 million independent observations would understate these standard errors by roughly the square root of the average cluster size; the 1.27 million owner clusters, not the filings, carry the information. The final row of the upper panel drops Madrid §66(a) registrations altogether rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger, not smaller, without them.

Three properties of Table 3 matter. The gradient is monotone within cells (+0.26, +0.52, +0.98, +2.29pp across quintiles two through five before controls). The full control set cuts the magnitude by about a third but does not absorb it, so it is not absorbed by who files or how they draft. It is, if anything, *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose gate failure rates are high for their own reasons.[5] And a large share of the raw pooled contrast is composition: the design-based

[4]Ties matter more here than they usually do. About 2.7% of scored filings share a ΔKL value with another filing, and the largest tie group runs to fourteen. Ties were far commoner under class-year reference windows, where boilerplate text scored against a shared reference produced identical values by the thousand; anchoring the window on the filing date all but removes them. Quintile cut points are therefore fixed by sorting on the score and then on the serial number, so the assignment is reproducible. Splitting a tie group across a boundary remains arbitrary; re-estimating under a rank rule that keeps tied filings together, at the cost of unequal quintiles, moves the top-versus-bottom contrast by at most 0.03pp in the pooled and stratum specifications, and by up to 0.21pp in the thinner cohort thirds.

[5]The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp. The Madrid figure is a substantive one now that §71 cancellations are read alongside §8: those registrations fail this gate more often than domestic ones. Reading §8 alone put the same coefficient at −46pp, which was an artifact of scoring them as survivors.

magnitude is +1.4pp against a raw +2.4pp, a reduction of about forty percent.

The penalty is modern. It is indistinguishable from zero for the 2002–2005 cohorts, turns positive in 2006, and runs +1.8 to +5.4pp for every cohort from 2008 onward (Figure 4).

That timing invites an out-of-sample check, and the 2019 cohort supplies one: it is the first whose gate window falls entirely after the pandemic, and it reproduces the recent-cohort magnitude at +5.5pp raw among fully-elapsed registrations ($n = 313{,}725$). The 2020–2021 gates have not yet elapsed at the data edge.

The filer population also changed over exactly this period, which is why the cohort thirds of Table 3 matter. *Within* counsel-represented US-domiciled owners, and with the full control set, the earliest third is indistinguishable from zero and in fact slightly negative (−0.25pp, SE 0.33), while the two later thirds are clearly positive (+2.00pp and +2.65pp). The leading gate risk is therefore a feature of the post-2008 filing economy itself, not of the post-2015 surge in self-filed and foreign applications, and it predates the pandemic by a decade. That the effect is absent rather than merely smaller before 2008 is a stronger statement than the annual-bucket version of this table supported, where the same third read +1.04pp.

Levels and gradients separate cleanly across filer strata. Self-filers fail the gate far more often than represented owners (69.6% versus 46.8%), and intent-to-use registrations more often than use-based ones (53.1% versus 50.2%), but the ΔKL gradient itself is similar across these strata once class, cohort, and drafting covariates are held fixed (+1.8pp self versus +1.2pp counseled in Table 3): who files predicts the level of gate risk; what the filing says predicts the increment on top of that level. Per class, the largest penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beverages, transport and construction.

The risk persists at the year-ten renewal, and on the design-based estimates does not attenuate: the lead coefficient is −0.49pp at the first gate and −0.63pp at the second (Table 6). What attenuates is the raw single-class contrast. In the software diagnostic class, where the event-dated first-gate lift on the 2010–2013 cohorts reaches +12.1pp, the conditional lift at the second gate falls to +7.6pp; pooled across classes the second-gate pattern is a shallow U (40.5% at the bottom quintile, 37.6% in the middle, 42.7% at the top), so by year ten the lagging tail has closed the gap in cumulative failure — marks whose language their industry was drifting away from end up abandoned at much the same cumulative rate as the marks it moved toward, having simply taken longer to get there. Most of the difference resolves at first proof of use. Response timing in the prosecution record sharpens the picture: among represented software filings, response speed to office actions predicts the gate (fastest-quintile responders pass at 57.4% against 49.4% for the slowest), and the penalty survives within both fast and slow halves (−7.2 and −8.0pp), so the gradient is not an artifact of already-failing firms being detected late: it is present in both response-speed halves. This particular split is computed on *term* scoring under annual reference buckets rather than the per-filing windows used elsewhere. Term scores exist only in the class-year construction, so every term-scored quantity in this paper is on annual references; among the topic-scored outcome analyses the exceptions are this split and the window decomposition below. No topic-scored version of this split exists, so it should be read as corroborating the direction of the result and not as one of its magnitudes.

Mixing the past and future reference windows identifies which kind of novelty the gate punishes. The penalty loads on breaking with the established past (−7.3pp) rather than on resembling the future (−3.2pp), and in software the separation is complete (−12.7pp against −0.3pp). This decomposition is the other topic-scored analysis on annual reference buckets rather than per-filing ones — because it requires a separate scoring at each of $W \in \{3, 5, 7\}$ on both sides; by the evidence of Appendix A.5 its magnitudes are likely inflated by something like a third, and only the contrast between the two configurations is being relied on here. Appendix A reports the full decomposition, plus a look-ahead-free months-scale variant that reproduces the pattern and a snapshot cross-check that agrees with the event-dated estimates under every scoring tried.

4.3 Being early costs at the public-reporting margin; being unusual does not pay there

Panel B of Figure 1 contains two patterns, and they pull in different directions. The outcome throughout this subsection is whether an owner ever appears in the Securities and Exchange Commission's financial reporting records — its EDGAR filing system and its published company list. That marks a firm as SEC-reporting, which is a broader and coarser category than exchange-listed: roughly half the matched firms carry an exchange ticker and the remainder report for other reasons. It rises monotonically in the KL *levels* on both axes (from 0.19% at the bottom past-facing quintile to 0.52% at the top in raw bins). And in signed ΔKL it is U-shaped: both tails beat the middle quintiles, with the lagging tail highest (0.44% at Q1, 0.24% at Q3, 0.39% at Q5).

Table 4: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
Atypicality quintiles, $A = \frac{1}{2}(K^{-} + K^{+})$ (Q1 omitted)		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^{-} alone	−0.003	0.018
Signed ΔKL quintiles, FE + log length + KL levels		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
Decomposition, FE + log length		
|ΔKL| (per σ)	+0.067	0.009
signed ΔKL (per σ)	−0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 3.

The design-based estimates (Table 4) resolve both patterns into one decomposition, and it is the raw atypicality gradient that does not survive it. Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by −0.003pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition — firms that go on to report cluster in particular classes, years, and multi-class filings — and class×year fixed effects absorb it. Lead costs, at equal atypicality: the signed component enters negatively (−0.047pp per σ, $t = -7.8$) while the unsigned component enters positively (+0.067pp per σ, $t = 7.6$). The U-shape in raw bins is exactly this pair: lagging filings show the atypicality premium with no lead penalty, leading filings show it net of that penalty, and the middle quintiles show neither. Once the levels are controlled, the apparent recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and flattens rather than recovering (Table 4 middle panel).

How owners are linked to public-market records governs what this result can mean, so the construction is worth stating. Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. Restricting the candidate universe — to a subset of classes, say — makes matchability itself a function of the regressor, and the atypicality gradient is sensitive to exactly that. Its absence here is therefore the more credible reading of the two, and the signed axis, which does not move when the linkage is varied, carries the weight. The financing rows of Table 6 rest on a Form D match whose raw extracts are not in the current tree and should be read as provisional.

The two axes do different work, and only one of them survives at the firm level. Atypicality — writing about a product in language unusual for its category — predicts survival at the use-proof gate (Section 4.2), where marks written in category-typical language are abandoned more often. It does not predict which owners become SEC-reporting companies once composition is absorbed. Lead — writing in language the category subsequently adopted — is the timing bet, and it costs at both margins: at the gate, and at SEC reporting even holding atypicality fixed. Being early is expensive; being unusual is not rewarded here.

The two are not independent in the obvious economic sense, which is worth stating because it is the natural reading of the pair. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language is one that now describes its goods the way you describe yours. Being early therefore consumes the very quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything, but because the category arrived. On this reading atypicality is not a fixed endowment but a position that competitors erode, and lead measures how fast the erosion happened. The design cannot demonstrate that mechanism — it cannot separate a category catching up from a common cause moving both — but it is what a negative coefficient on lead at equal atypicality would look like if the mechanism were real.

4.4 The construct is not patenting

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), a two-way fixed-effects regression of standardized firm-year mean ΔKL on $\log(1 + \text{patents})$ gives -0.014σ ($t = -2.3$) at $T = 50$ and -0.010 ($t = -1.5$) at $T = 200$ — essentially uncorrelated within firm, which is the property the construct needs.

Table 5 reports the more hostile version of the same comparison: it scores ΔKL under an exponential-decay reference kernel on terms rather than the flat window on topics used everywhere else in this paper, and it returns a larger negative coefficient, -0.048σ ($t = -7.9$), on 174,569 firm-years (94,106 with within-firm variation across 14,470 firms). Its strongest negative sits on patents_{t-1} (-0.054, SE 0.009), which would say that in years after a firm patents heavily its trademark vocabulary regresses toward class-typical — temporal substitution between patent prosecution and market-facing brand work. That reading should be held loosely. The lags are entered jointly and $\log(1+\text{patents})$ is strongly autocorrelated within firm, so the split between the contemporaneous and lagged coefficients is partly arithmetic; and under the paper's own scoring, with each timing entered separately, the lag is not distinguishable from zero (-0.009, $t = -1.08$; the full set is reported below). The orthogonality claim rests on the topic-scored estimates, which are the ones the current pipeline reproduces.

Two objections to reading that null as orthogonality are worth taking seriously, and both fail. The first is timing. Under a linear model of the innovation process a patent is applied for before its invention is in public use, while a trademark is filed when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly, however, and in some industries the trademark application comes first (Seip et al., 2018), so the ordering is an empirical question rather than an assumption to build on. Entering patents at $t-2$, $t-1$, t, and $t+1$ — the last allowing the trademark to precede the patent — leaves nothing at any timing: -0.015 at two years out ($t = -1.61$), -0.009 at the lead year ($t = -1.08$), -0.010 contemporaneous ($t = -1.51$), and $+0.003$ with the trademark first. No ordering recovers a co-movement.

The second objection is heterogeneity: a pooled near-zero is consistent with genuine orthogonality everywhere, but equally with a positive slope in sectors where the two rights are complements — pharmaceuticals, chemicals, medical devices, machinery — offsetting a zero or negative slope where trademarks do the work alone. Splitting the panel on a complement/neutral/substitute grouping fixed in advance — assigned by judgement from the class definitions rather than estimated, with the eleven classes covering chemicals, pharmaceuticals, machinery, electronics, medical apparatus and scientific services treated as complements, the consumer-goods and services classes as substitutes, and the remainder neutral — and re-estimating the identical specification inside each cell, gives -0.000 ($t = -0.03$) among complements, -0.026 ($t = -1.45$) among substitutes, and an interaction of -0.008 ($t = -0.49$). Ordering all 27 Nice classes that clear a 50-owner floor by coefficient, the complement classes run from sixth place to twenty-fourth and do not agree in sign with one another: pharmaceuticals $+0.011$, software and electronics $+0.003$, medical devices -0.002, chemicals -0.070, machinery -0.090. The classes that do reach significance run against the complementarity story rather than merely failing to support it — the only significant positives are transport and cosmetics, neither of which is a complement class, while the significant negatives include medical devices and machinery, both of which are complement classes.

This sample makes the test harder rather than easier. It is conditional on a successful assignee match, so every owner in it patents at least once; the complement and substitute cells therefore both contain patenting firms and resemble each other more than the industries they stand for do. That pushes any interaction toward zero. The weight consequently rests on the cell-level nulls rather than on the interaction test — and these are the firms among whom a patent–vocabulary relationship should be easiest to find, since all of them patent.

4.5 Each axis bites at a different stage

The outcomes above are estimated in separate places on separate samples, which obscures the pattern that matters most: the two axes do their work at different points in a mark's life, and where each one bites is informative. Table 6 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner's first filing — under a single specification, so the coefficients can be read down a column. Its magnitudes differ from the section-specific tables above, each of which uses its own sample and specification; the signs and the ordering are what the table is for.

Table 5: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit $\log(1 + \text{patents})$. *This table alone scores ΔKL on terms under an exponential-decay reference kernel*, not on topics under the flat window used throughout the rest of the paper; the topic-scored estimates quoted in the text are smaller and are the ones the claim rests on. Lead/lag coefficients are from a single joint regression, so they are not independent of one another.

Specification	coef (σ)	SE
Between-firm correlation (firm means)	−0.020	—
Within-firm correlation (firm+year demeaned)	−0.015	—
Two-way FE, contemporaneous	−0.048	0.006
robustness: patents = max over matched assignees	−0.048	0.006
controlling log(n_filings)	−0.051	0.006
Lead/lag: patents$_{t-1}$	−0.054	0.009
patents$_t$	−0.019	0.010
patents$_{t+1}$	+0.010	0.010

Standard errors in place of significance stars; see the note to Table 3.

Two reversals stand out, and both turn on the difference between the examiner and the market. Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (−1.35pp per standard deviation), then survive the first use-proof gate more often (+1.35pp) and reach public markets more often (+0.04pp on a 0.56% base — the one specification in which a positive atypicality coefficient on listing survives, and it is a continuous term through a non-monotone profile, not the quintile contrast of Table 4). The examination process penalizes surprise; the market does not. Having been early runs the other way: leading applications complete registration *more* often (+0.87pp) and then pass the use-proof gate *less* often (−0.49pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on a sixth of the sample: atypicality +1.30pp and lead −0.63pp among marks that had already cleared the first gate. Whatever the first gate is selecting on, the second gate selects on it again rather than reversing it. Those filers follow through on the paperwork; what does not follow through is the product.

The registration row needs one caution. A linear coefficient cannot represent the inverse-U documented in Section 4, where both vocabulary tails complete registration less often than the middle. The positive linear term reflects the asymmetry of that shape — the leading tail completes more often than the lagging tail — and not a monotone gain from leading.

The financing rows are the ones worth dwelling on, because atypicality does nothing at one stage and a great deal at the next. Whether a debut owner ever raises a Regulation D round — a private securities placement, reported to the SEC on Form D — is unrelated to how atypical its filing was: −0.017pp per standard deviation among owners who completed registration, against a 2.61% base, which is indistinguishable from zero ($t = -0.8$). Among owners who did raise one, the same measure predicts *reaching* the public market: +0.49pp on a 3.49% base, about 14% relative ($t = 3.8$). Atypical language is invisible at the financing stage and informative afterwards, which is what an investor screen that does not read the text would look like — the text being free and public at the moment of filing. The comparison is among survivors, so it cannot separate that reading from selection on unobservables; and the null at entry is a null, not evidence of indifference.[6]

That last row requires one construction that is easy to get wrong. An exchange-registration

[6] This pair is sensitive to the reference window. Under annual reference buckets the entry row was −0.06pp ($t = -2.8$), which read as active negative selection. Per-filing windows move it to zero and leave the listing row intact, so the negative selection was an artifact and the informativeness among the funded was not.

Table 6: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner's first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference. Cohort windows differ because each maintenance gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The first gate is a dated cancellation for non-use; the second is whether a registration that cleared the first was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten death share a terminal status and are separable only this way. This table alone is estimated at $T = 200$ topics rather than the $T = 50$ used elsewhere; the sign and ordering of every row are unchanged at $T = 50$.

				Unsigned		Signed
Outcome	Given	n	Base	Atypicality	$\lvert\Delta\text{KL}\rvert$	Lead
Reached registration	filed	3,161,552	56.11%	−1.352 (0.033)	−0.307 (0.044)	+0.872 (0.028)
Passed first gate (yr 6)	registered	1,260,392	41.82%	+1.349 (0.054)	−1.150 (0.082)	−0.487 (0.052)
Passed second gate (yr 10)	passed first	297,944	55.92%	+1.296 (0.108)	−0.297 (0.162)	−0.629 (0.105)
Raised a Reg D round	filed	1,535,004	2.53%	+0.003 (0.015)	+0.148 (0.027)	−0.149 (0.016)
Raised a Reg D round	registered	916,448	2.61%	−0.017 (0.020)	+0.115 (0.035)	−0.114 (0.021)
Listed after the round	funded	38,886	3.49%	+0.493 (0.129)	+0.608 (0.170)	−0.513 (0.104)
Became SEC-reporting	registered	1,773,938	0.56%	+0.042 (0.007)	+0.040 (0.011)	−0.004 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 3.

marker — the filing a company makes with the SEC to list its shares on an exchange — on its own does not mean a firm listed *after* raising: 56% of matched owners carrying one registered before their first Regulation D notice, because established public companies also place securities privately. Read naively, the row would be measuring public companies doing private placements rather than funded firms going public. It is therefore restricted to owners whose exchange registration postdates their first notice, which cuts the positive coefficient roughly in half from its uncorrected value and leaves it significant.

Three further limits bound how hard the reading can be pushed. Regulation D notices are electronically filed only from 2009, so the financing rows rest on 2009–2018 debut cohorts rather than the full panel. Owners are matched to filers by normalized name, which succeeds for a small and unrepresentative minority and favours formally constituted entities, so every financing estimate is conditional on matchability. And the final row conditions on funding, which is realized after the filing being scored; the coefficient is a contrast among survivors, not a treatment effect. What it establishes is that information distinguishing eventual listers from other funded firms was already present in the goods/services text years earlier — not that atypical language caused the outcome.

Lead, by contrast, does not reverse anywhere after registration. It is negative at both use-proof gates, at financing with or without registration imposed, and at listing among the funded; the only row where it is not negative is SEC reporting among registered owners, where it is indistinguishable from zero (−0.004pp, SE 0.007). Being early is penalized at every stage where the market, rather than the examiner, is doing the selecting, and never rewarded at any of them.

5 Business vocabulary flows out of software, and entrants do not carry it disproportionately

The spatial version of this question is answered: von Graevenitz et al. (2022) show that novel description tokens reach distant US locations more slowly than near ones, in the *Regional Studies* programme surveyed by Castaldi & Mendonça (2022). The cross-industry version is open, and the Nice class system supplies the partition.

Cross-industry *vocabulary* diffusion is directly measurable on this corpus. For nine curated business-model phrases, Table 7 records the first year each of four classes (software 009, tech services 042, transport 039, advertising/retail 035; 2.32 million granted filings) crosses a 1% prevalence floor. Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size, and the asymmetry is a four-class result a 45-class build would test properly.

Table 7: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Unsupervised theme extraction corroborates the pattern without depending on which phrases were picked. A detection-purpose latent Dirichlet allocation (LDA) topic model ($T = 50$ on a stratified 200,000-filing sample) yields recognizable business-model components; for each (theme, class) cell crossing the 1% floor with six or more tracked years, Bass-model fits (Bass, 1969) on 118 retained cells give median innovation coefficient $p = 0.018$, median imitation coefficient $q = 0.110$, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Tagging each multi-class theme's origin class and counting directed origin-to-arrival edges: software contributes 33 of 72 edges against 18 expected under symmetry, and receives only 10. The asymmetry reproduces under independently-fitted non-negative matrix factorization (NMF) on the same sample; both are bag-of-words factorizations, so the robustness is within-paradigm.

Entrants carry the canonically new themes, but no more often than they carry anything else. The 50 earliest carriers per theme are 73.7% entrants on average — but the corpus's year-matched debut baseline is 76.4%, so the theme-carrier excess is −2.7pp and 28 of 50 themes sit *below* baseline. The themes leaning incumbent relative to baseline include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing: entrants are no over-represented among the earliest carriers of a new theme than among filers generally, so the arrival of new vocabulary is not an entrant phenomenon in the way the Schumpeterian reading would predict (Schumpeter, 1942). This base-rate discipline is general — population-share questions on this corpus mislead unless the debut rate is netted out.

The AI era, monitored this way, has so far compounded rather than erupted — but the most recent data are testing that reading rather than confirming it. Aligning era starts (internet 1994, AI 2015) and tracking era-term prevalence in the software and tech-services classes: through year three the two track closely (5.1% versus 4.0% of filings), then diverge — internet vocabulary tripled between years four and six, reaching 23% of filings in 2000, while AI vocabulary reaches 12.9% at year nine, roughly where the internet stood at year five (Figure 5). The growth is not uniform, and the last two observed years are the steepest in the series: prevalence rises about 14% a year from 2019 to 2021, then 50% between 2022 and 2023 and a further 26% into 2024, following the release of consumer chatbots. That is an acceleration rather than a discontinuity — per-year corpus turbulence shows no AI-era deviation through 2024 — but it runs in the direction the eruption reading predicts, and two more cohorts would distinguish the two paths. Combined with the incumbent lean of the AI theme, the record through 2024 still reads as incumbent-led vocabulary adoption rather than an entrepreneurial explosion. The comparison is sensitive to the era-start alignment chosen, and the falsifiable marker is now close to binding: if AI follows the internet's path with a lag, its prevalence should triple within roughly three years of the 2024 data edge, and the entrant share of its carriers should rise above the corpus baseline rather than sit below it.

6 What the measure supports, and what it does not

Four claims are defended. ΔKL, adapted from Barron et al. (2018); Murdock et al. (2017) onto topic distributions of USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. Vocabulary position carries two separable signals: atypicality marks marks that survive their use-proof gate (+1.3pp per standard deviation), and lead marks product risk (+1.4pp top-versus-bottom quintile at that gate with fixed effects, controls, and owner clustering; −0.047pp per σ in SEC reporting at equal atypicality). Atypicality does *not* predict SEC reporting once class and year are absorbed, and an earlier reading of these data that said otherwise rested on a narrower owner-to-registrant linkage. The construct is empirically distinct from patent-track invention within firm. And the corpus supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and asymmetric flow out of software on the four classes built so far.

Three readings are rejected. ΔKL does not measure innovation in any strong sense; the defended primitive is vocabulary position on commercial text, and Appendix E shows concretely which kinds of novelty the primary scoring cannot see. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic. And the term-scored signed firm-level advantages of leading filings (margins, listing) are artifacts of drafting style and lexical specificity (Appendix B).

6.1 What the measure is not: five neighbouring constructs

Novelty and *transience*, in the sense of Murdock et al. (2017) and Barron et al. (2018), are the two KL levels themselves: novelty is divergence from the past, transience divergence from the future. Atypicality as used here is their average, and so is neither of them alone; *novelty* in particular is declined because it names only the backward-looking half of a quantity that faces both ways. What does not carry over at all is the interpretation. In those corpora one author writes the document and the reference stream is that author's own intellectual environment, so a high value says something about the author's cognitive state. Here each filing is scored against other firms' filings, so the same number says only that the description sits far from what its industry was writing.

Resonance is their signed difference, and is arithmetically what this paper calls lead. The word is avoided because it names a mechanism — the field resonating with an author's contribution — that this design cannot observe. A filing can sit where its class was heading because it was imitated, because it and the class responded to the same outside event, or by coincidence; the measure does not separate these, and *resonance* would assert the first.

Innovation is a property of the product or service, not of the sentence describing it. Two firms selling the same thing can draft very differently, and a genuinely new product can be described in boilerplate; Appendix E shows a filing whose invention is encoded entirely in ordinary words. The measure sees prose.

Radicalness, in the innovation-management sense (Dewar & Dutton, 1986; Henderson & Clark, 1990; Tushman & Anderson, 1986), is the magnitude of a departure from established practice, with the implication that existing competences are devalued. It is the closest of these neighbours to what atypicality captures, since KL is a departure measure and not a newness count — the Amazon exhibit is a large departure built from unremarkable components. It is still declined, because radicalness is a claim about the underlying technology or business model, and nothing here licenses the step from unusual language to devalued competence.

Distinctiveness has two prior occupants and is avoided for both. In trademark law it is the registrability doctrine, proven in part by the continued use this paper uses as an outcome. In strategic management, optimal distinctiveness (Zhao et al., 2017) predicts an inverted-U in conformity — a shape that appears here at registration but not at listing, so borrowing the term would imply the theory is being tested when it is not.

What is left, and what *atypicality* is meant to say, is narrower than any of them: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

One dependence in the data is worth stating because it bears on how any descriptive display of this corpus should be read. Filings are not independent draws: some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.42 for gate survival and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The estimates above are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. But binning filings without that correction overstates how much structure the corpus contains: on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

The main open questions are dated and specific. Why the gate penalty emerged with the 2008–2010 cohorts is unresolved: the within-stratum estimates rule out the post-2015 filing-surge composition story, but cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot yet be separated. A prospective-only variant of the construct, with reference vocabularies and topic bases refit on vintage data, would close the remaining look-ahead channel (the current representation is fit on the pooled corpus; the months-scale freshness variant is already look-ahead-free and reproduces the risk pattern in its diagnostic class). Beyond these: linking gate failure to observable commercial death (owner filing cessation, EDGAR exit); the 45-class diffusion build; theme-value portability across industries; the supplier-versus-deployer split within arriving themes; demographic enclaves and theme origination; validation against the USPTO ID Manual's codification history; and a cross-paradigm (embeddingbased) theme extraction.

7 Reproducibility

Code, firm-year panels, and analysis outputs are available at `https://github.com/ericsilver/tm-vocabulary`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the underlying TRTYRAP backfile is approximately 12 GB streamed once). The published panels include the SEC-matched firm-year ΔKL panel (n = 59,033) and the PatentsView-joined panel (n = 174,569). Analysis scripts and their outputs are indexed in the repository `README.md`.

An online appendix at `https://github.com/ericsilver/tm-vocabulary/tree/master/docs/online-appendix` carries the class-level results behind every pooled estimate reported here: a three-panel breakout for each of the 45 Nice classes, the cross-industry comparisons, and a machine-readable table of per-class contrasts with standard errors.

A Computation, corpus, and robustness

A.1 Corpus and scoring detail

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The topic representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-topic distribution, and prospective and retrospective KL are computed against per-filing topic references spanning 1,826 days on each side of the filing date, scoring 1995–2019. A term-level variant is retained alongside: token distributions (unigram + bigram, within-class document frequency ≥ 50) against class-year windows with Dirichlet smoothing ($\alpha = 1$), computed at $W \in \{3, 5, 7\}$. The term variant is used where vocabulary identity is the object (phrase transit, the window decomposition, the freshness measure) and as the comparison scoring throughout.

A.2 Burn-in and interpretable range

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 6). The burn-in cut is set empirically: thin past references inflate prospective surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of ∼2.0 nats — the natural-log unit in which these divergences are measured — in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations (∼0.15–0.17) are not burn-in — mechanical contamination cannot rebound — but the dot-com era moving the corpus. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

A.3 Reliability and fragilities

Past-facing and future-facing KL are highly correlated however scored: on the complete class-009 corpus $r = +0.971$ under term scoring and $+0.979$ under topic scoring at $T = 200$ ($n =$ 1,107,807), the latter stable across resolutions (0.980 at $T = 50$, 0.977 at $T = 500$). The lead L is therefore a small residual of two large, stable quantities: under the per-filing windows used throughout, both levels have a standard deviation near 0.75 nats about a mean of 1.94, while L has a standard deviation of 0.125 — about a sixth of either level's spread, and under a fifteenth of the mean level. Only about 1.4% of the two levels' combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels, and under term scoring the dominant such thing is document length. Term-scored atypicality correlates with log filing length at -0.729 in class 009 and -0.688 in class 035: the shorter the description, the spikier its term distribution and the further it sits from any smoothed reference, so "unusual" and "short" are very nearly the same measurement. Topic scoring breaks that identity, taking the same correlations to -0.251 and -0.125. The consequence for the lead is direct: the standard deviation of L rises threefold across deciles of A under term scoring and by a quarter to a half under topic scoring, so a unit of lead means roughly the same thing everywhere only in the topic representation. Appendix D shows this as a length surface over the two axes. Two qualifications belong with those numbers. Topic scoring returns a finite value for about 65% of filings, so the two columns are not the same sample; re-estimating the term-scored correlations on exactly the topic-scored subset still gives -0.722 and -0.692, which makes the contrast a property of the representation rather than of the sample, but the like-for-like spread comparison is closer to twofold against 1.3–1.6-fold than to the headline threefold. And "flat" means no monotone length gradient rather than constant: class 035 under topic scoring is mildly U-shaped in A. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored observations), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Appendix B). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit $+0.09\sigma$ above the panel mean under term scoring ($p = 0.22$), $+0.14\sigma$ under topic scoring ($p = 0.07$), and $+0.08\sigma$ at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with ΔKL on the same panel (Pearson +0.010).

A.4 Scoring robustness of the mark-level results

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.4pp survival across quintiles under topic scoring and -5.7pp under term scoring on the identical sample (Figure 7). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp prospective, +7.6pp retrospective): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 4.3, at the mark level.

A.5 What annual reference buckets cost

Every estimate in this paper uses a reference anchored on the filing's own date. The alternative is one reference per class-year: the class's topic distributions were averaged over the five preceding calendar years, and every filing made in year t was scored against that single object, so a filing made on 1 January and one made on 31 December received identical references and neither saw any of the language filed during its own year. Because the whole corpus has been scored both ways, the cost of that approximation can be stated rather than speculated about.

It leaves a signature in the data. Under annual bucketing, mean lead in the software class rises almost monotonically with filing month, from -0.006 in January to $+0.013$ in December — an eighth of a standard deviation produced by nothing but position within the calendar year, since a December filing sits eleven months further from its past reference and eleven months nearer its future one. Per-filing windows remove it: the month-to-month spread falls from 0.020 to 0.005 and loses its monotone shape. The two scorings agree only moderately. On 1,107,807 software filings scored both ways the signed axis correlates at $r = 0.67$ and the levels at 0.77, and the per-filing version is the tighter, with a standard deviation of 0.125 against 0.160.

Three consequences for the results, in decreasing order of importance.

First, the gate penalty was overstated by about a third. On the same 3.10 million registrations, the top-versus-bottom contrast is +2.15pp (SE 0.30) under annual buckets and +1.39pp (SE 0.17) under per-filing windows. The attenuation is remarkably uniform across specifications — the ratio runs 0.59 to 0.73 for every sample split in Table 3 — and it is an attenuation of magnitude only: standard errors fall by roughly half, so the t-statistic *rises*, from 7.2 to 8.3. The same holds in the software class alone, where the contrast moves from +7.40pp to +5.25pp.

Second, the shape of the gate penalty changes. Under annual buckets the quintile profile looked like a gradient with a steeper final step; under per-filing windows it is close to a threshold, with the top quintile carrying about seventy percent of the total and the interior quintiles nearly flat. The coarser measure spread a top-quintile effect across the whole range.

Third, one published shape does not survive. Registration completion showed an inverse-U on both the unsigned and the signed axis. Only the unsigned one is real; on ΔKL the curvature disappears entirely under per-filing windows, and the earlier interior peak is attributable to the month gradient above. The firm-level results are the least affected by the window change, though they are sensitive to the owner-to-registrant linkage instead (Section 4.3): the signed decomposition keeps both its signs and its significance under either window.

Nothing here changes a sign or a conclusion. It changes magnitudes, one functional form, and the confidence with which the magnitudes can be quoted.

A.6 Which novelty: the reference-window decomposition

"Leading" is relative to a reference, and mixing windows across the two sides identifies which novelty the gate punishes. Scoring the corpus at symmetric windows of $W \in \{3, 5, 7\}$ *years* — this appendix measures the window in years, where the main text measures it in days — plus a foresight configuration (prospective against the recent 3-year past, retrospective against the distant 7-year future) and a past-rupture configuration (the reverse): the top-versus-bottom gate lift runs -3.2pp for foresight, -4.7pp at symmetric $W = 3$, -5.7pp at the $W = 5$ baseline, -6.1pp at $W = 7$, and -7.3pp for past-rupture, the most pervasive variant (negative in 35 of 41 classes; Figure 8). In software the decomposition is complete: foresight carries no penalty (-0.3pp) while past-rupture carries -12.7pp. Use-proof-gate risk attaches to breaking with the established past, not to resembling the future, consistent with vocabulary leaving the corpus gradually: the long-past reference holds the fading establishment, and distance from it is what the use requirement punishes. The registration inverse-U is indifferent to the window mix (+4.6 to +4.9pp depth across all five variants), so the examiner and the marketplace respond to different aspects of the same text.

The decomposition extends below the year. Scoring class 009 monthly against two adjacent references (the twelve months ending one month before filing, and the twelve before those), the two levels are nearly identical per filing ($r = 0.992$) but their difference — a months-scale freshness measure, high when a filing resembles the very recent months far more than the year before them — reproduces the risk pattern at -9.4pp across quintiles, stronger than the symmetric baseline in the same class, with the penalty on the high-freshness tail — filings whose language runs ahead of the recent months rather than behind them — which is the same direction as the annual-window result. Freshness is built entirely from pre-filing text and is computable on the filing date with no look-ahead; corpus-wide estimation is owed.

B The measurement hazard: term scoring manufactures firm-level signal

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per σ ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with prospective surprise entering positively and retrospective negatively; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per σ at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed topic-ΔKL is flat to declining, and the margin coefficient reverses: -1.3pp per σ ($t = -1.8$) at $T = 50$, -2.2pp ($t = -3.1$) at $T = 200$, and -0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the prospective/retrospective decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The topic model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, topic scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only ∼9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed ΔKL gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (-6.9 and -8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-ΔKL mechanically. Topic count is a partition dial, not a coverage dial — a crypto/blockchain topic exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

The portable conclusion: term-resolved text-novelty measures can manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic. Four completions are owed: rescoring topics on the full per-class term vocabulary; a future-blind vintage refit of the topic model; decomposing term-ΔKL into in-vocabulary and out-of-vocabulary contributions and re-estimating the firm panels on each; and an emergenceevent design comparing carriers of to-be-emergent terms against class-, year-, and length-matched controls with attorney-of-record fixed effects.

C The outcome analysis without registration conditioning

Figure 9 repeats the lifecycle outcome analysis without the registration conditioning (term-scored, where the composite shape was originally encountered), on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is an inverse-U in ΔKL (quintiles 22.0%, 22.9%, 23.0%, 22.5%, 21.4%; middle-versus-tails depth +1.4pp, SE 0.1pp), because the composite inherits the registration step's middle-favoring shape from Figure 1A. Unconditional survival analyses on this corpus therefore recover an inverse-U whether or not the maintenance gate itself favors the middle, which is why the body of the paper conditions on registration throughout. Panel B shows the conditional gate outcome on the same filings for contrast; on this mixed one-to-two-gate cohort it is mildly inverse-U with a clearly penalized leading tail, consistent in the right tail with the cleaner single-gate cohort of Figure 7 and a reminder that the left tail of the gate result is cohort-sensitive.

D What each representation measures

Figure 10 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under topic scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns, so the same document can be watched moving between representations. Apple's iPod (2001) has a lead of +0.972 under term scoring and +0.065 under topic scoring; Amazon.com (1997) goes from +0.763 to −0.011. Both read as pronounced leaders on terms and as unremarkable on topics, which is the firm-level hazard of Appendix B in miniature.

This appendix and the next cut in opposite directions. Topic scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which topic scoring does not, and the same Amazon filing serves as the exhibit in both places. The paper reports topic scoring for every estimate, because the length confound contaminates firm-level comparisons, and term scoring for every worked example, because the topic basis has no coordinate for the language the examples turn on.

E How three filings encode

Table 8 traces three filings through term scoring and topic scoring at $T \in \{50, 200, 500\}$, illustrating three distinct kinds of novelty and how each representation handles them.

Amazon.com (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the surviving bigram "line search" is the act of forming the compound caught in progress. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under topic scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

KOMBUCHA (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the topic vocabulary but, with no kombucha-like topic, its mass dissolves into a generic beverages topic (≥ 0.60 weight at every T).

ETHEREUM (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services topics at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain topic — the eleven occurrences of *software* outvote the blockchain terms.

Table 8: Top topic loading for three filings, by representation. Term scoring resolves the novel content of all three; topic scoring resolves none of it at any tested resolution.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

The pattern across the row is the measurement point in miniature: topic scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting topic exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses topic scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Appendix B).

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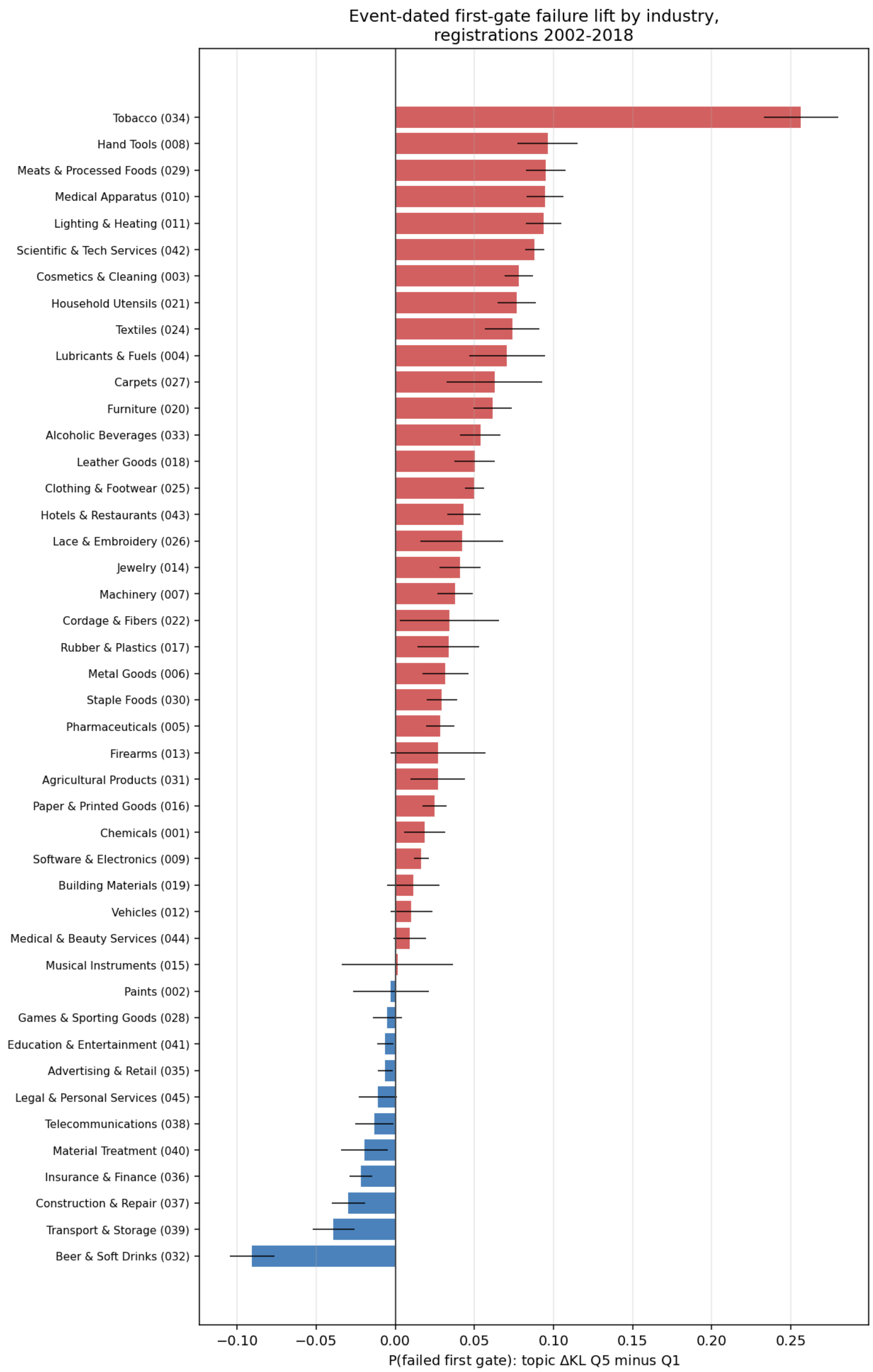

Figure 3: Event-dated first-gate failure lift (topic ΔKL top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals and do *not* correct for the within-owner correlation of gate outcomes (0.42), so they are narrower than clustered intervals would be; the design-based estimates in Table 3 are the ones to read for precision. The online appendix gives a three-panel breakout for each class and the cross-industry comparisons behind this figure.

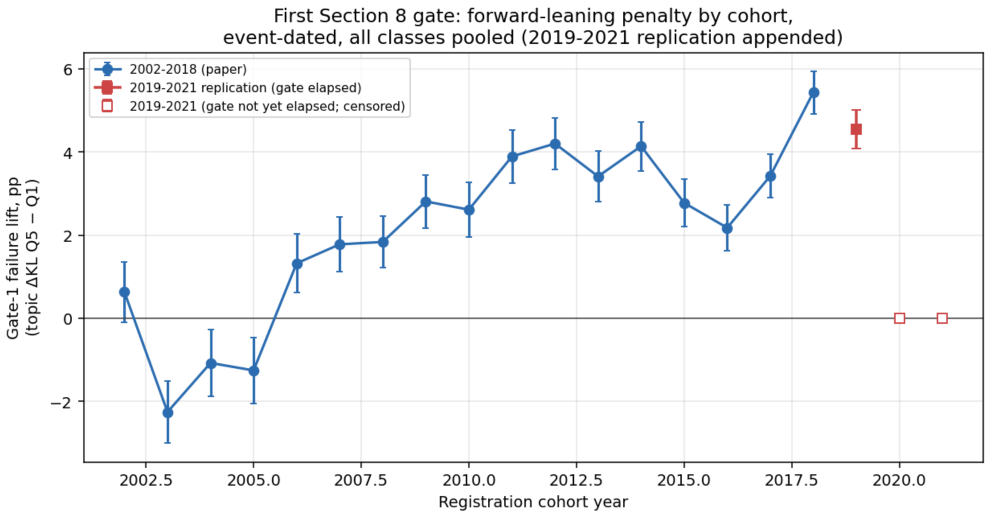

Figure 4: The leading gate penalty by registration cohort, event-dated, all classes pooled, raw quintile contrasts. The penalty is absent or negative for the 2002–2005 cohorts, turns positive in 2006, and is present in every cohort from 2008 to 2018, ranging +1.8 to +5.4pp with no downward trend. The 2019 point is the out-of-sample replication on fully-elapsed registrations (+5.5pp); the 2020–2021 gates have not yet elapsed.

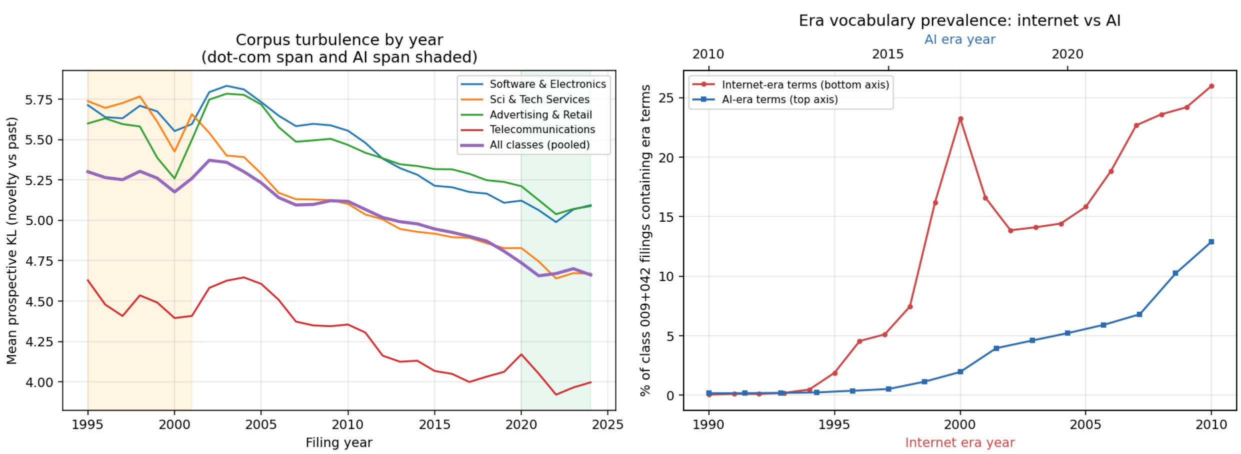

Figure 5: *Left:* per-year mean prospective KL by class and pooled, 1995–2024, with the dot-com and AI spans shaded. *Right:* era-term prevalence in classes 009+042, internet era (bottom axis, from 1994) against AI era (top axis, from 2015).

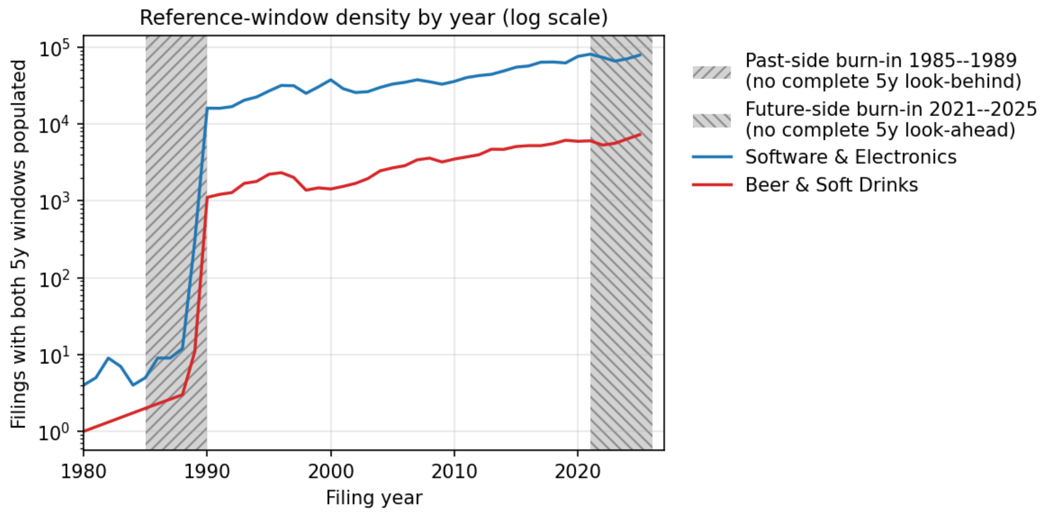

Figure 6: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

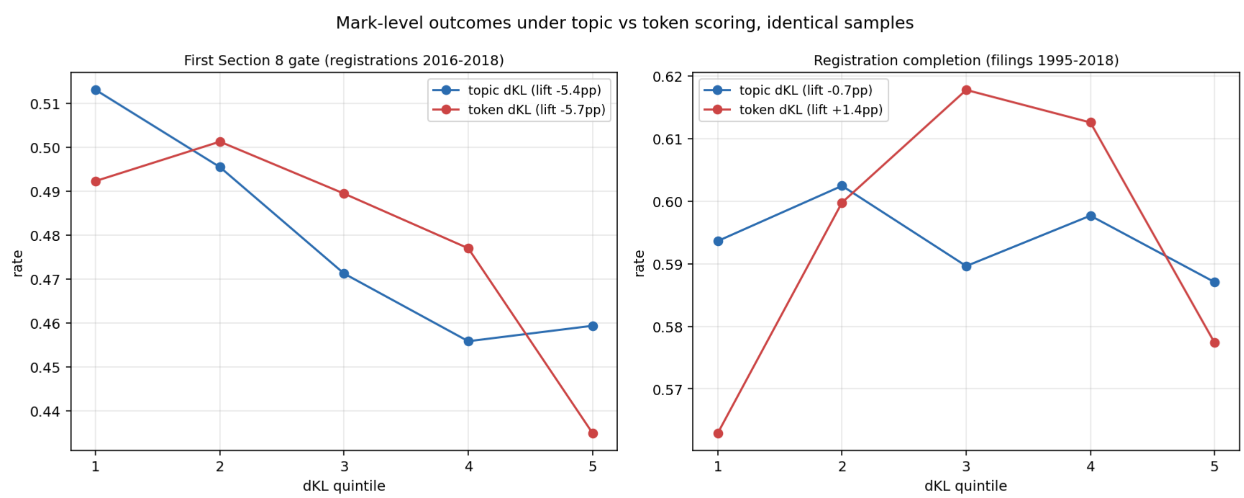

Figure 7: The two mark-level outcomes under topic and term scoring on identical samples, pooled across classes. *Left:* first use-proof gate survival (snapshot, registrations 2016–2018), where the two representations agree closely. *Right:* registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under topic scoring (depth −0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and Section 4 reports only the unsigned version.

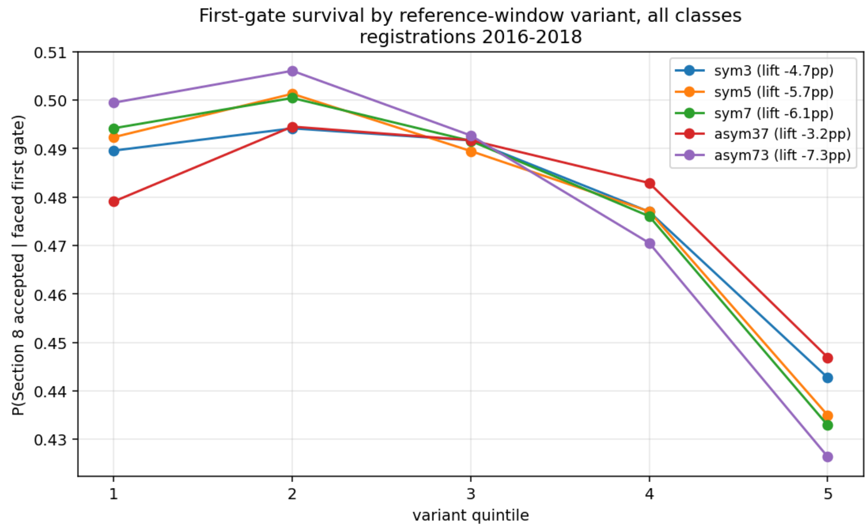

Figure 8: First-gate survival by quintile of each reference-window variant, pooled across classes, registrations 2016–2018. The penalty deepens monotonically from the foresight configuration (pros W=3, retr W=7) to the past-rupture configuration (pros W=7, retr W=3).

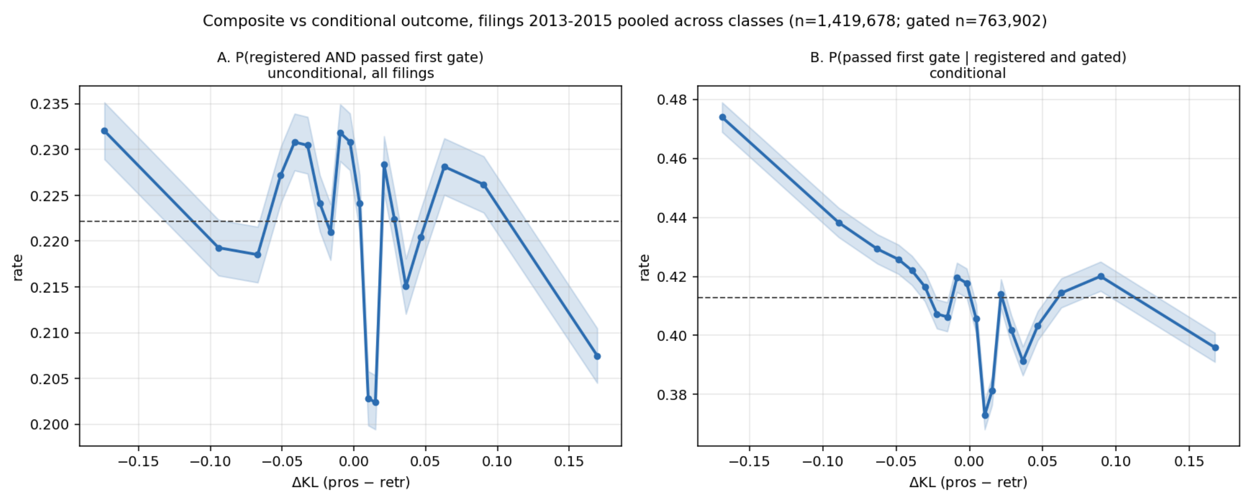

Figure 9: Filings 2013–2015 pooled across classes, topic-scored on per-filing reference windows (n = 1,419,678; gated subset n = 763,902). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

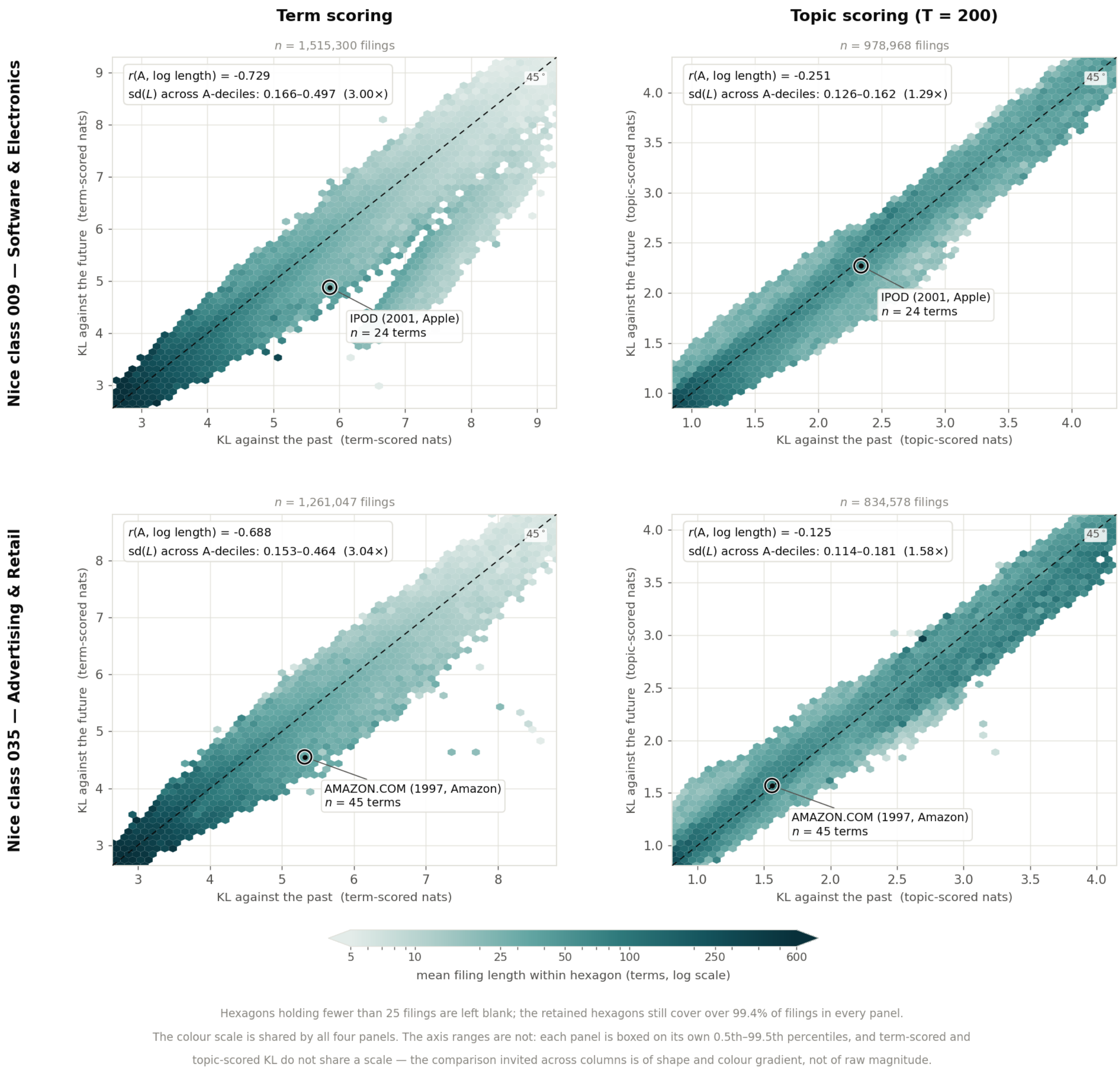

Figure 10: Mean filing length across the two axes, by representation and class. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and topic-scored KL do not share a scale, so the comparison invited across columns is of shape and colour gradient rather than of magnitude. Topic scoring returns a finite value for roughly 65% of filings, so the columns are not the same sample; the correlations quoted in Appendix A are re-estimated on the common subset and are substantially unchanged.